Mohammadali Sefidi Esfahani

Master of Science in Computer Science from Amirkabir University of Technology

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Education

PhD in Software Engineering *Sept. 2024 - April 2026*

Concordia University *Montreal, Canada*

- **Supervisor:** Professor Peter C. Rigby
- **Research**: SE knowledge management systems and the impact of developer turnover
- **Withdraw due to not receiving my study visa**

Master of Science in Computer Science (Soft Computing and Artificial Intelligence) *Sept. 2021 - Sept. 2023*

Amirkabir University of Technology (AUT) *Tehran, Iran*

- **Cumulative GPA:** 19.62 out of 20
- **Master's Thesis:** Identification of Unspecified Events from Twitter
- **Supervisor:** Dr. Mohammad Akbari
- *Graduated as a 1st-ranked student*

Bachelor of Science in Computer Science *Sept. 2017 - July 2021*

Kharazmi University (KHU) *Tehran, Iran*

- **Cumulative GPA:** 18.26 out of 20
- **Bachelor's Thesis:** Comparative Study on the Performance of ML Models for Fall Detection in Older People
- **Supervisor:** Dr. Mohammad Soltanian
- *Graduated as a 1st-ranked student*

Honors and Awards

- Concordia University Tuition Award of Excellence valued at $48888, Gina Cody School of Engineering and Computer Science, Concordia University, 2026 (declined due to visa delays)
- Concordia University Graduate Doctoral Fellowship valued at $56000, Gina Cody School of Engineering and Computer Science, Concordia University, 2025 (declined due to visa delays)
- Concordia University Split Graduate Fellowship GCS valued at $28000, Gina Cody School of Engineering and Computer Science, Concordia University, 2024 (declined due to visa delays)
- Ranked 1st among the MSc. computer science students (2021 entrance), Department of Mathematics and Computer Science, Amirkabir University of Technology
- Ranked 1st among the BSc. computer science students (2017 entrance), Department of Mathematics and Computer Science, Kharazmi University

Publications

Mitigating the Risk of Defects and Improving Knowledge Distribution with Code Reviewer Recommenders

Mohammadali Sefidi Esfahani & Peter C. Rigby, **Accepted in FSE'26**, Montreal, Canada, 2026

- **Link:** FSE'26 Website

The Cost vs the Benefit of Adding an Extra Code Reviewer to Mitigate Developer Turnover through Reviewer Recommenders

Mohammadali Sefidi Esfahani & Fahimeh Hajari & Peter C. Rigby, **Presented at ICSE'26**, Rio, Brazil, 2026

- **Link:** ICSE'26 Website

EnrichEvent: Enriching Social Data with Contextual Information for Emerging Event Extraction

Mohammadali Sefidi Esfahani & Mohammad Akbari, Iran Journal of Computer Science, 2025

- **Link:** Springer

Comparative Study on Performance of ML Models for Fall Detection in Older People

Mohammadali Sefidi Esfahani & Mohammad Fattahian, Journal of Artificial Intelligence, Virtual Reality, and Human-Centered Computing, 2025

- **Link:** AI-VR-HCC Journal

Feature Learning for Identification of Persian Events from Twitter

Mohammadali Sefidi Esfahani & Mohammad Akbari, CSICC'23, 2023

- **Note:** This paper is written in the Persian Language
- **Link:** Civilica & PDF

Experience

Graduate Research Associate at Concordia University *Sept. 2024 - April 2026*

Supervisor: Professor Peter C. Rigby

- **Duties:** Research on SE knowledge management systems and the impact of developer turnover.

Graduate Research Assistant in Data Science Innovation Center *Nov. 2021 - Sept. 2024*

Supervisor: Dr. Mohammad Akbari

- **Duties:** Research, Management of Data Science Innovation Center weekly meetings, Mentoring undergraduate members

Teaching Assistant (Machine Learning) *Sept. 2023 - Feb. 2024*

Lecturer: Dr. Ali Mohades Khorasani

- **Duties:** Programming assignments, Grades of programming assignments, Troubleshooting sessions

Lead Teaching Assistant (Machine Learning) *Sept. 2022 - Feb. 2023*

Lecturer: Dr. Mohammad Akbari

- **Duties:** Assignments, Grades of assignments and final project, Troubleshooting sessions

Graduate Research Assistant in Natural Language Processing Innovation Center (NLPIC) *Nov. 2021 - Aug. 2022*

Supervisors: Dr. Ali Mohades & Dr. Mohammad Akbari & Dr. Mohammad Ebrahim Shiri

- **Duties:** Research

Member of ACM association *March 2021 - Aug. 2021*

Supervisor: Dr. Bardia Panahbehagh

- **Duties:** Event planing and management

Projects

Sofia-ExpertPlusPlus (GitHub)

Replication Package for FSE'26 Paper

- In this paper, we design reviewer recommendation strategies to mitigate the risk of defects and improve knowledge distribution.

Sofia-LearnerPlus (GitHub)

Replication Package for ICSE'26 Paper

- In this paper, we investigate the cost vs. benefit of assigning an extra reviewer to pull requests with knowledge at risk to turnover.

EnrichEvent (GitHub)

Replication Package for my Master Thesis

- An end-to-end framework for identifying unknown events from Twitter in an unsupervised setting.

Comparative Study of ML Models for Fall Detection in Older People (GitHub)

Replication Package for my Bachelor Thesis

- This project aimed to implement and evaluate the efficiency of diverse ML models for fall detection in older people.

No-Show Appointments Prediction

Real-World Project (Private-Access)

- This project aimed to implement an end-to-end framework for no-show appointment prediction based on patient history.

Data Generator for Comparative Opinion Summarization using Chat-GPT (GitHub)

Data Science Innovation Center Project

- The implemented tool scrapes users' reviews for hotels from Eghamat24.com website and builds a synthetic dataset for comparative opinion summarization.

Small Projects in AI (GitHub) / Small GNN-based Projects (GitHub)

Self-Study

- Collection of small projects in various fields like ML, Data Mining, Deep Learning, NLP, and GNNs

Research Interest

- Code Reviewer Recommendation
- AI and ML for Software Engineering
- Explainable AI
- Computational Social Science
- Applied ML for Industrial Systems
- Representation Learning

Skills

Programming Languages Python, Java, C-sharp
Libraries and Frameworks TensorFlow, Pytorch, Pytorch-Geometric, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, dotnet
Tools Git, Latex, Adobe Photoshop, Search Engine Optimization (SEO)

Languages

English TOEFL iBT (Overall: 98/ Reading: 24/ Listening: 26/ Speaking: 22/ Writing: 26/ Test Date: 18 November 2023)
English MSRT (Overall: 75/ Listening: 25/ Structure & Written Expression: 21/ Reading Comprehension: 29/ Test Date: 26 September 2025)
Persian Native

Certificates

Deep Learning Specialization (Credential link) *Nov. 2023*
Issued by Coursera: DeepLearning.AI
- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models

Natural Language Processing Specialization (Credential link) *Nov. 2023*
Issued by Coursera: DeepLearning.AI
- Natural Language Processing with Classification and Vector Spaces
- Natural Language Processing with Probabilistic Models
- Natural Language Processing with Sequence Models
- Natural Language Processing with Attention Models

Machine Learning Specialization (Credential link) *Oct. 2023*
Issued by Coursera: DeepLearning.AI and Standford
- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, Reinforcement Learning

Task-oriented course in artificial intelligence and machine learning (Credential link) *Nov. 2021*
Issued by Quera

Hobbies

- Futsal
- Volleyball
- Photography
- Camping
- Ping-Pong
- Swimming
- Mountain Climbing